

30-SECOND P0 IMPLEMENTATION SUMMARY

P0-ALL-STOP-01 · NOT_AUTHORIZED · HOLD

P0-CAND-01 | 12 tasks | 17 roles | Layer A 8/8 PASS | 4-FTE working roster | non-formal CAPEX/OPEX | field + 12 gates HOLD

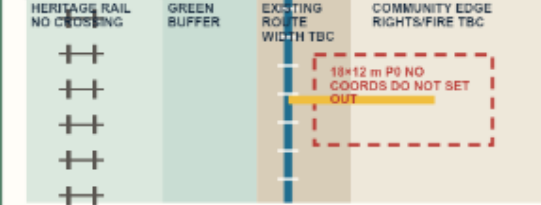
Any group safety-critical failure, missing equivalent, or failed exit => whole unit HOLD

FIXED REVIEW FIGURE 03 / v1.4 DELIVERY CONTROL

P0-ALL-STOP-01 | DIMENSIONED LAUNCH UNIT

P0-CAND-01 · NOT_AUTHORIZED · HOLD Provisional-area
screening relation · no coordinates/set-out

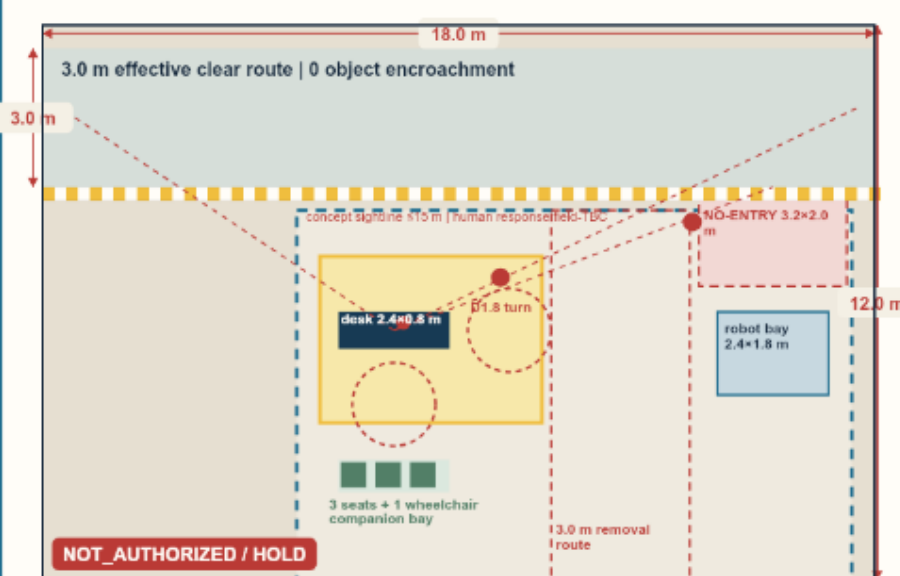
1:500 SITE RELATION | CANDIDATE SCREENING ONLY



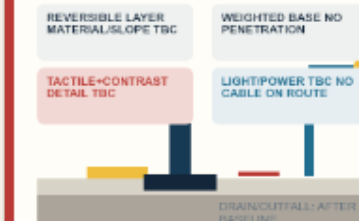
1:50 SECTION | CLEARANCES, INTERFACES, REVERSIBLE COMPONENTS



1:100 CONDITIONAL PLAN | ASSUMPTIONS; MOVE / SHRINK / PAUSE / REMOVE



1:20 KEY INTERFACE | REVERSIBLE



DIMENSIONS | ASSUMPTIONS ONLY

D01	18x12 m	sheltering envelope
D02	12x8 m	reversible ground
D03	3.0 m	effective clear width
D04	01.8 m	turning space
D05	2.4 m	desk length
D06	015 m	concept sightline
D07	2.4x1.8 m	robot bay
D08	3.2x2.0 m	no-entry zone
D09	0.6 m	object setback
D10	1.5 m	work clearance
D11	3.0 m	removal route
D12	17.28 m²	shelter area

TBC: tactile/contrast, light, drainage, emergency width, capacity

FAILURE CONDITION → AUDITABLE ACTION

MOVE	rights / heritage / fire / utility / drainage conflict	screen-out + new candidate baseline
SHRINK	only if clear route, equivalence, maintenance, exit stay intact	recalculate dimensions / BOQ / roster
PAUSE	roster, audit, weather, light, maintenance, complaint gap	stop log + human takeover + restart decision
REMOVE	critical safety failure, missing equivalence, failed exit, group rejection	incident pack + T12 removal/restoration
RESTORE	day 90 or any removal instruction	before/after + closure + acceptance

THE SLOW LINE

P0 DIMENSIONS, SECTION + INTERFACE

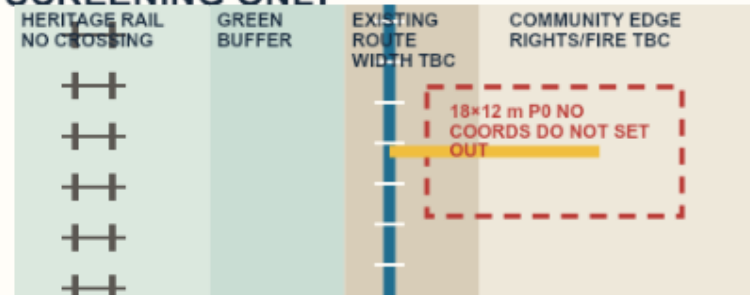
A3 · v1.4 · 02

FIXED REVIEW FIGURE 03 / v1.4 DELIVERY CONTROL

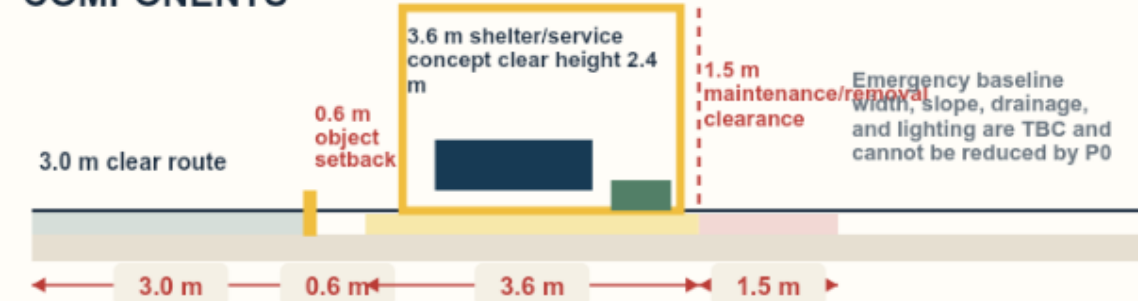
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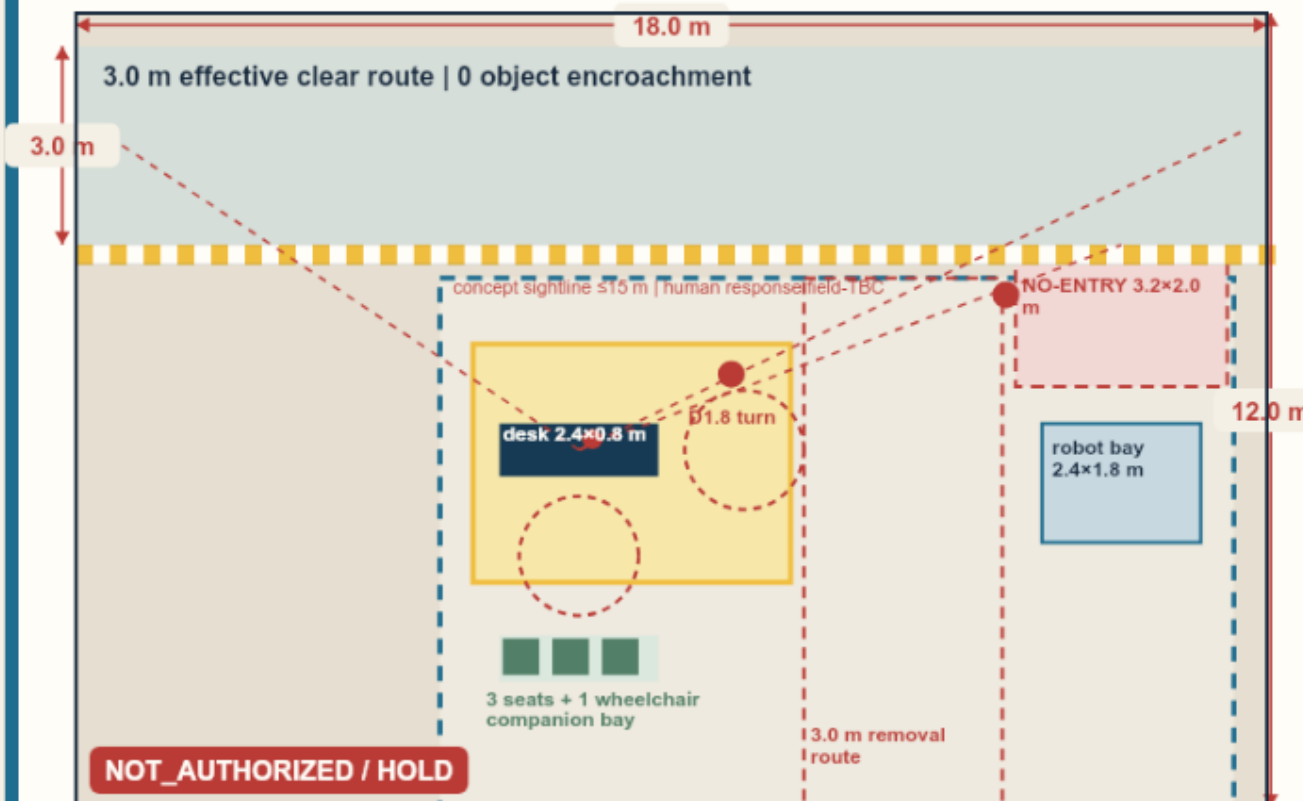
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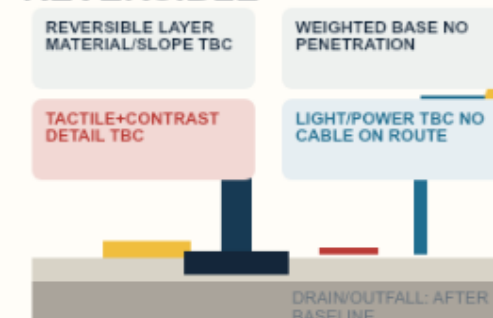
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D02	12x8 m	reversible ground
D03	3.0 m	effective clear width
D04	D1.8 m	turning space
D05	2.4 m	desk length
D06	≤15 m	concept sightline
D07	2.4x1.8 m	robot bay
D08	3.2x2.0 m	no-entry zone
D09	0.6 m	object setback
D10	1.5 m	work clearance
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THE SLOW LINE

A3 · v1.4 · 03

TASKS, ACCOUNTABILITY + SIX GATES

P0-ALL-STOP-01 | D00-D90 | 12 TASKS | 17 ROLE SLOTS | 12 EXTERNAL GATES HOLD | NOT AUTHORIZED

90-DAY DELIVERY CHAIN | TIME, RESPONSIBILITY, GATE + EXIT EVIDENCE



ACCOUNTABILITY HAND-OFF + IMMEDIATE STOP POWER | ALL 17 SLOTS unassigned/conditional

R-P0-EXEC P0 implementation coordinator	A-P0-RIGHTS Site-rights holder/commissioning accountability slot	R-P0-CODESIGN Paid accessibility co-design lead	R-P0-SERVICE Staffed-service and human-takeover operator	R-P0-SAFETY On-duty safety/privacy professional	R-P0-INSTALL Demountable works installation/restoration role
R-P0-EVAL Independent accessibility/operations evaluator	R-P0-SURVEY Survey and baseline-record professional	R-P0-ACCESS Accessibility professional reviewer	R-P0-TRAFFIC Traffic and slow-mobility interface reviewer	R-P0-FIRE Fire and emergency-route reviewer	R-P0-STRUCTURE Structural and temporary-works reviewer
R-P0-ELECTRICAL Electrical, lighting, and isolation reviewer	R-P0-DRAINAGE Surface, level, and drainage reviewer	R-P0-EQUIPMENT Equipment stop, rescue, and interface reviewer	R-P0-LIGHTING Lighting and visual-readability reviewer	R-P0-MAINTENANCE Maintenance, spares, and reinstatement role	

Full input, output, HOLD, and recovery/exit evidence remains in the delivery register.



Any user or worker may activate the physical stop without penalty; G0-G5 remain 6/6 default closed.

THE SLOW LINE

NON-PRICED QUANTITIES, COST + TWO-LAYER ACCEPTANCE

A3 · v1.4 · 04

16 BOQ LINES | 4-FTE WORKING ROSTER | NON-FORMAL CAPEX/OPEX | LAYER A 8 PASS / 0 HOLD | B 12 HOLD

NON-PRICED BOQ | 16 LINES TRACE TO PLAN, SECTION OR TASK

P0-Q01	Frame	1 set	P0-Q09	Robot stop line	3.2 m
P0-Q02	Shelter	17.3 m²	P0-Q10	Robot no-entry marking	6.4 m²
P0-Q03	Reversible ground	96 m²	P0-Q11	Seating + companion bay	3+1 bay
P0-Q04	Staffed desk	1 desk	P0-Q12	Lighting points	4 points
P0-Q05	E-stop facilities	2 units	P0-Q13	Interface cabinet + points	1+2 pts
P0-Q06	Paper rack	1 rack	P0-Q14	Install + four checks	1+4 checks
P0-Q07	Wayfinding points	5 points	P0-Q15	Maintenance visits	13 visits
P0-Q08	Tactile / contrast edge	18 m	P0-Q16	Remove + restore	1 lot

Market rates, quotation entity, currency, formal total and basis date: null / TBC

TWO-LAYER ACCEPTANCE | A: 8 PASS / 0 HOLD | B: 12 HOLD

A01	Route connectivity	PASS
A02	Obstacle/encroachment count	PASS
A03	Non-AI equivalent coverage	PASS
A04	Default Gate state	PASS
A05	Audit-record completeness	PASS
A06	Malformed input triggers HOLD	PASS
A07	Human task remains complete with AI off	PASS
A08	Exit/restoration process completeness	PASS

LAYER B REQUIRES FIELD BASELINES: wheelchair, low vision, no-smartphone time to staffed help, human response, flow conflict, noise, lighting, drainage, microclimate, resident acceptance, roster and actual cost.

AUDIT FIELDS (BY metric_id): formula → data source → threshold state → responsible role → current status → trigger
Synthetic tasks, fictional journeys and general opinions cannot substitute for Layer-B field performance.

Any group safety-critical failure, missing equivalent service or failed exit => whole unit HOLD; no averaging away failure.

COST + ROSTER WORKING BANDS | FORMAL ESTIMATE/QUOTE NULL

C_P0 = REV + HUMAN + CO-DESIGN + ACCESS/SAFETY
+ PRIVACY/EVAL + O&M + REMOVE/RESTORE + RESERVE

REV	Reversible space works	PRIVACY_EVAL	Privacy and independent evaluation
HUMAN	Staffed service	OM	Operations and maintenance
CODESIGN	Paid co-design	REMOVE_RESTORE	Removal and site restoration
ACCESS_SAFETY	Accessibility and safety review	RESERVE	Contingency/restoration reserve

CAPEX CNY 0.85–2.10m | OPEX 0.90–2.20m/year | 12 h: 3.129 → 4 FTE | participant sensitivity only

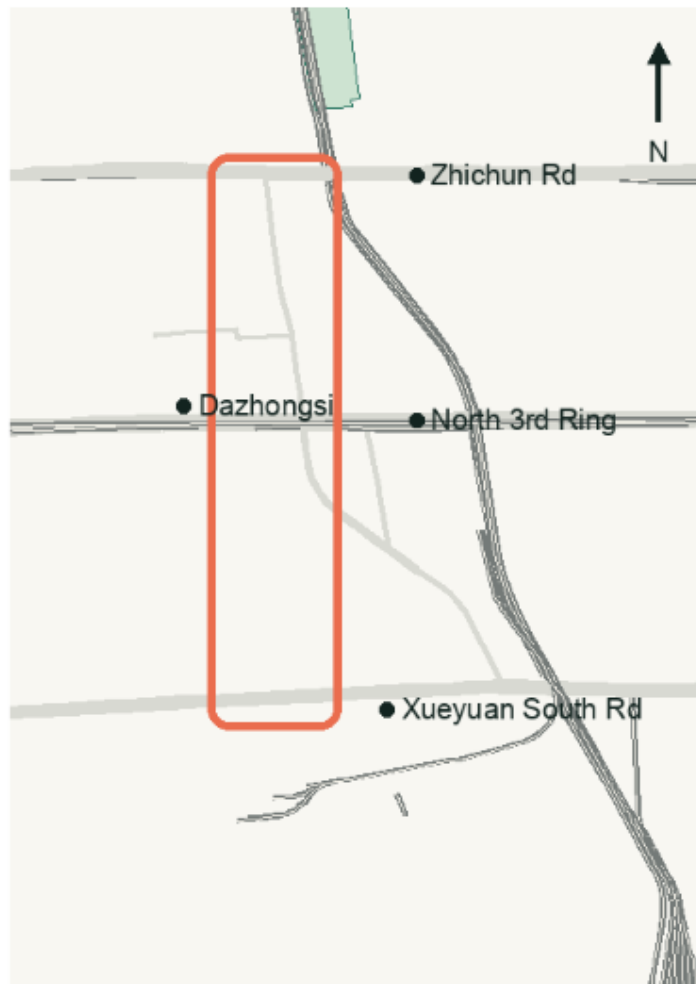
FORMAL null

SITE GROUNDING / REPRESENTATIVE SEGMENT

VERIFY ORIENTATION, THEN ADD THE DESIGN

Representative segment | orientation + text-only facts + reversible overlay

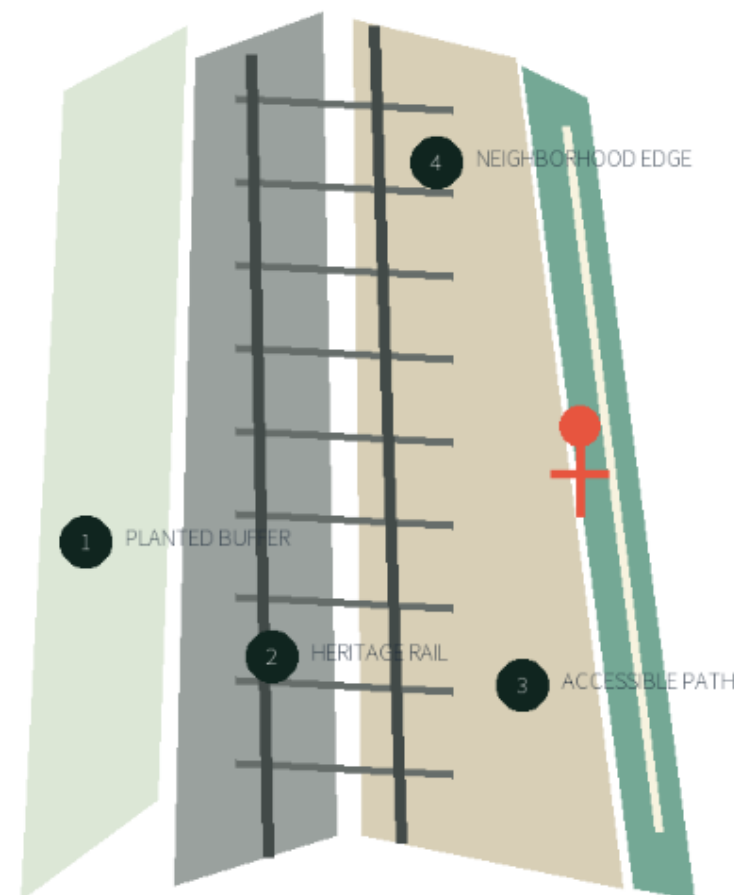
A ACTUAL ORIENTATION



OSM orientation base. Orange band marks the representative study segment—not a red line or exact site.

B TEXT-ONLY WORKING SKELETON

Published facts translated into a diagram



NOT AN EXISTING-CONDITIONS SURVEY.
Orientation and published text only.

C REVERSIBLE OVERLAY | NOT APPROVED

KEEP

rail corridor · planted buffer · public route

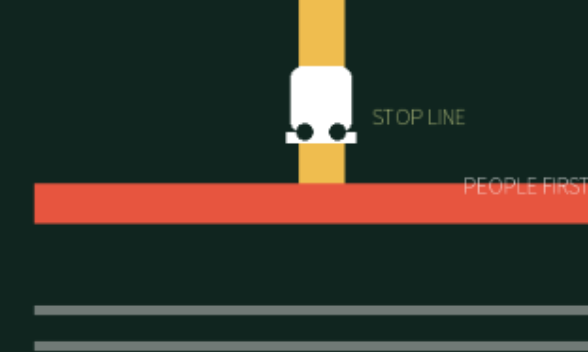
ADD

staffed kiosk · physical E-stop · side spur

BAN

robot on path/rails · automated first launch

PERPENDICULAR SERVICE SPUR



Actual siting still requires survey, tenure, mobility, heritage and operating approval.

Orientation/data: © OpenStreetMap contributors (ODbL), snapshot 2026-08-29. Text facts: Beijing Municipal Government report, 2026-07-30. Not a survey.

THE SLOW LINE

ONE LINE, THREE YARDS, SIX ALL-STOP GATES

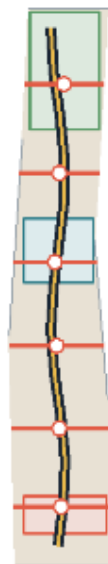
A3 · v1.4 · 06

THE SLOW LINE | Keep pace with the slowest person

Not slower innovation: a city latency budget for safety, comprehension and choice

01

ONE LINE · THREE YARDS · SIX ALL-STOP GATES



SAFE-SPEED YARD

NO-LOGIN COMMONS

EVERY-STOP MARKET

Dashed = repository provisional envelope, not the real park alignment or planning boundary

SLOWEST-USER TEST · SUT-6

- MOBILITY · wheelchairs & slow walkers
- SENSORY · low vision & hearing loss
- DIGITAL · no phone, no account
- LANGUAGE · children & newcomers
- CARE · carers & strollers
- TIME · night and shift workers

EVERY-STOP CONTRACT

purpose → slowest user → non-AI equivalent → human takeover → minimum data → physical stop → appeal → public review

FROM STRATEGY TO GROUND: THREE LEVELS, ONE DECISION CHAIN

43.6 km² research → 11.4 km² provisional container → three transferable prototypes

02

43.6 km²

RESEARCH · put slowest first into industry, talent and governance

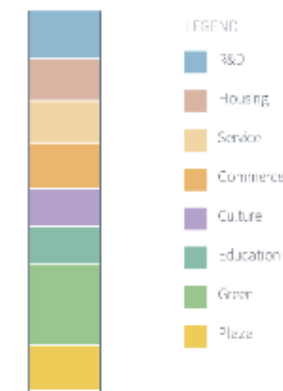
11.4 km²

OVERALL DESIGN · one Slow Line and six all-stop gates

3 AREAS

KEY DESIGN · Safe Speed, No Login, Every Stop

CONCEPT LAND USE | RECOMPUTABLE



Concept only · replacement official controls arrive

ONE MAP: SPACE × SERVICE × OPERATIONS

SPACE

- continuous accessible line
- six cross-city gates
- three reversible yards

SERVICE

- works without login
- a human is findable
- non-AI path is equivalent

OPERATE

- pilot before building
- failure defaults to stop
- publish review before scale

Official geometry → replace constraints → regenerate all layers, metrics, figures and PDFs

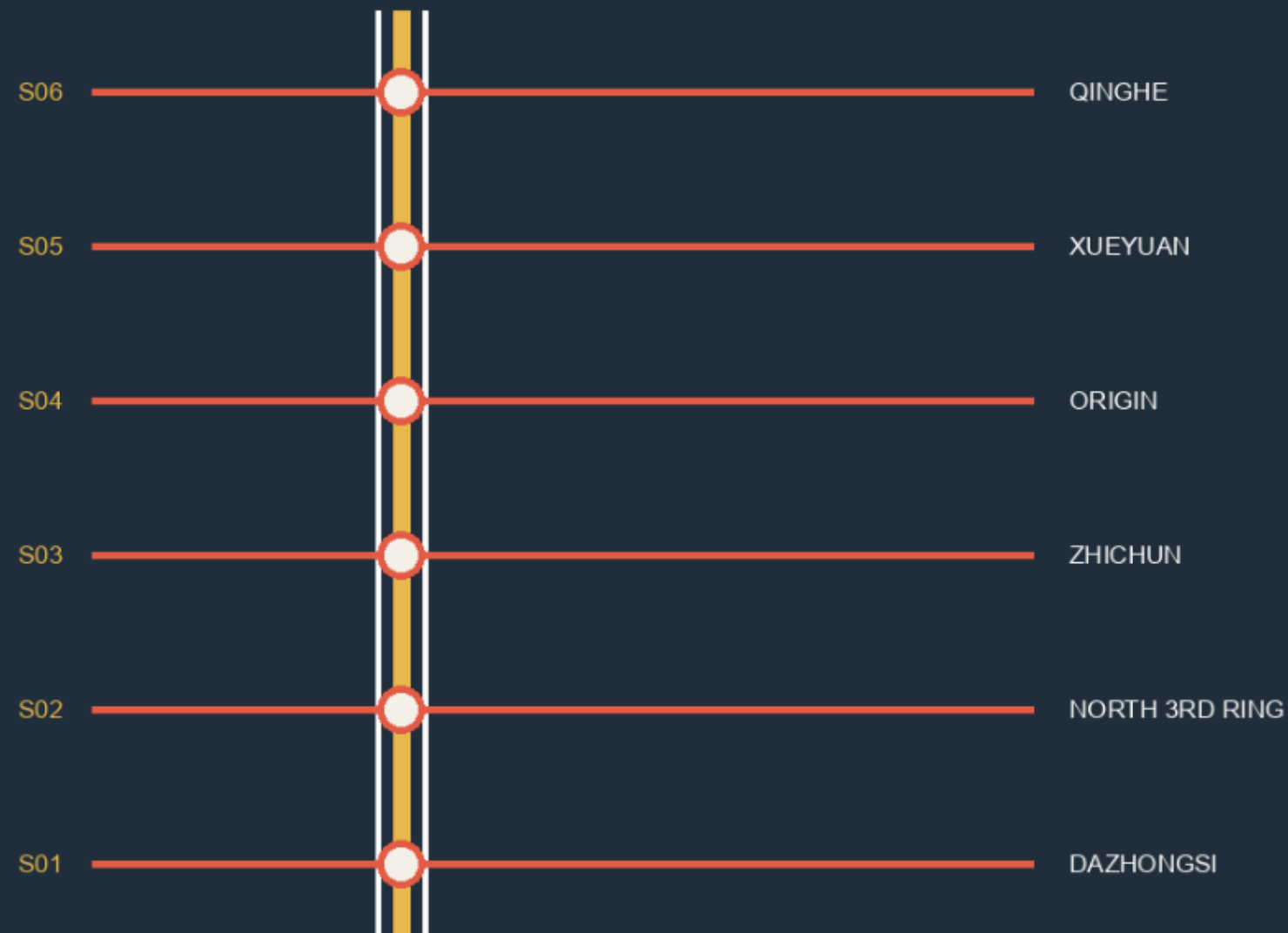


MOBILITY + BLUE-GREEN: THE FASTEST SYSTEM MUST WAIT

continuity, crossings, rest, help and stop authority before spectacle

04

CONCEPT NETWORK — NOT ROAD OR PARK ALIGNMENT



FIVE PERFORMANCE GATES

01 STEP-FREE CONTINUITY

detours do not count as access

02 REST + SHADE

spacing tested by slowest users

03 MULTI-SENSORY

visual, audio and tactile

04 HUMAN IS FINDABLE

on-site, phone and paper

05 ROBOTS CAN STOP

physical stop beats inference

NO SITE BASELINE: thresholds require co-design and professional review

OFFLINE REHEARSAL | PACKAGE CLOSED, ERRORS STILL BLOCK

OFFLINE REHEARSAL: PACKAGE CLOSED, ERRORS STILL BLOCK

Fixed seed | 12 synthetic tasks | no online model, people, personal data, or real robot

TASKS	REGISTERED RESULT	VALID PAYLOAD	MALFORMED REJECT	AUDIT COMPLETE	AI-OFF HUMAN EQUIV.
12	12/12	11/11	1/1	12/12	12/12

TASK LEDGER

SUT-001	PASS	SUT-002	PASS	SUT-003	PASS	SUT-004	PASS	SUT-005	PASS	SUT-006	PASS
SUT-007	PASS	SUT-008	PASS	SUT-009	PASS	SUT-010	PASS	SUT-011	PASS	SUT-012	HOLD OK

PACKAGE GAPS CLOSED IN v1.4

STAFFED DESK UNAVAILABLE Transfer to human telephone/text hot backup in the same service window; AI closes when both human paths are unavailable.	MALFORMED PAYLOAD Schemas remain invalid, but rejection must create a complete audited HOLD; the deliberate error is not painted green.	REAL-WORLD BOUNDARY 12/12 proves only fixed-ledger closure; any missing field baseline, appointment, funding, or permission remains HOLD.
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This proves only that the rule ledger is recomputable; it does not establish field accessibility, human response, robot performance, permission, or funding.

90-day All-Stop Gate: prove one pilot can stop

Three study windows · six evidence gates · six separately reported groups · NOT_AUTHORIZED

DAYS 0–30 Baseline + co-design Four-party walk · same task with AI off · barrier/dissent ledger Unclear tenure, fire status, or staffed equivalent → HOLD	DAYS 31–60 Closed offline rehearsal Emergency stop · human transfer · refusal · failure review Missing audit or AI open while staff desk closed → HOLD	DAYS 61–90 Time-limited micro-trial One time slot · one entrance · one bounded low-risk task Any group critical failure or failed restoration → HOLD
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SIX EVIDENCE GATES

G0 tenure/permission	G1 accessibility co-design	G2 staffed equivalent/roster
G3 privacy/safety/duty	G4 limited run/independent watch	G5 restore/public review

Same task, group by group

- Wheelchair users
- Blind/low-vision users
- Older people without smartphones
- Carers with children
- Non-Chinese speakers

One safety-critical failure, missing equivalent service, or failed exit in any group places the whole pilot on HOLD. No average can compensate.